Threshold-Aware Similarity Caching for Named Data Networks: Bounding Retry Delay under Congestion using ns-3

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Abstract

Named Data Networking (NDN) lets every router cache the Data packets it forwards, but, like any cache, it answers a request only with the exact object that was named [1]. Similarity caching relaxes this rule: on a miss, a cache may return the most similar object it holds, which suits image retrieval, recommendation and semantic caching of large language model queries [2], [3]. The user accepts the answer only if it is similar enough, and otherwise requests again. Making this work in a multi-hop network is difficult for four reasons: a cache does not know the similarity the user will accept, so it may send an answer certain to be rejected; NDN delivers Data only under the name of the pending Interest, so a substitute object cannot be returned as is; cache contents do not change within a retry interval, so a rejected user keeps getting the same answer; and the value of a nearby answer depends on upstream queueing and loss. Nakamura and Kamiyama [4] model the similar-content delivery delay on general cache networks, but assume a fixed answer probability α, lossless links of infinite bandwidth and retries that redraw cache contents independently, with numerical results only. Keeping contents fixed across retries on their topology, we find their model underestimates the mean delay up to 2.1 times at $\alpha = 0.5$ and up to 1700 times at $\alpha = 0.9$; other studies treat a single cache [2] or offline placement [5], not packet-level retries. We therefore propose threshold-aware similarity caching for NDN, in which Interests carry the user's minimum acceptable similarity and caches answer or forward based on the measured upstream delay. The objectives are: (i) to implement similarity caching with consumer retries in ns-3 using ndnSIM [6], reproduce the base paper on its tandem topology and measure the delay when cache contents persist across retries; (ii) to design threshold-carrying Interests with bounded retries and derive their delay in closed form; and (iii) to design a congestion- and loss-aware answer-or-forward rule and compare it with fixed-α caching under bottleneck links and burst loss on mean and 95th-percentile delay, delivered similarity and upstream traffic.

References

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